

Exercise: Inserting SQL

Context

You are working on a database for a small retail system that tracks **customers**, their **orders**, and related **payments**. Your task is to populate these tables using different forms of the `INSERT` statement.

Tables Overview

- **Customer**: Contains customer details
- **CustomerOrder**: Contains orders linked to customers
- **PaymentMethod**: Lookup table for payment types
- **Payment**: Records payments made for each order

Environment

1. Create a new PostgreSQL database named `retaildb`.
2. Run all setup scripts in this exercise against the `retaildb` database.

Setup Script (Run Before Starting)

Code:

<https://github.com/apprenti-org/software-developer-java-student/blob/main/databases-in-java/exercise-inserting-sql.sql>

SQL

```
CREATE TABLE Customer (  
    CustomerId SERIAL PRIMARY KEY,  
    CustomerName VARCHAR(100)  
);  
  
CREATE TABLE CustomerOrder (  
    OrderId SERIAL PRIMARY KEY,  
    CustomerId INT,  
    OrderDate DATE,  
    FOREIGN KEY (CustomerId) REFERENCES Customer(CustomerId)  
);  
  
CREATE TABLE PaymentMethod (  
    PaymentMethodId SERIAL PRIMARY KEY,  
    PaymentMethodName VARCHAR(100)
```

```

        PaymentMethodId INT PRIMARY KEY,
        MethodName VARCHAR(50)
    );

CREATE TABLE Payment (
    PaymentId SERIAL PRIMARY KEY,
    OrderId INT,
    PaymentMethodId INT,
    Amount DECIMAL(10, 2),
    FOREIGN KEY (OrderId) REFERENCES CustomerOrder(OrderId),
    FOREIGN KEY (PaymentMethodId) REFERENCES PaymentMethod(PaymentMethodI
d)
);

INSERT INTO PaymentMethod VALUES
(1, 'Credit Card'),
(2, 'Cash'),
(3, 'Mobile Payment');

```

Instructions

Part 1: Insert Single Records

1. Insert two new customers: *Emma Rivera* and *Noah Gray* into the **Customer** table.
2. Create an order for each customer with today's date in the **CustomerOrder** table.

Part 2: Insert with Auto-Increment Key

3. Insert a payment of \$49.99 for Emma's order using PaymentMethodId 1. Do not specify the **PaymentId**.

Part 3: Handle Foreign Keys

4. Try inserting a payment using **PaymentMethodId = 999**. What happens?
 - Write down the error message you receive.

Part 4: Bulk Insert

5. Insert three additional customers in a **single INSERT statement**:
 - Liam Davis
 - Olivia Brooks
 - Sophia Martinez

6. Insert a **bulk list of two orders** (use today's date) for any two of the new customers.

Reflection Questions

- What happens if you try to insert a record that violates a foreign key constraint?
- Why is using auto-increment helpful for primary keys?
- What are the benefits of bulk inserting records?